

Original article

Mortality rates of urban trees in relation to tree species and size development

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ABSTRACT

Urban trees provide vital benefits such as carbon storage and urban cooling. However, limited knowledge on urban tree mortality hinders effective management of sustainable urban tree stock. Utilizing cadaster datasets, this study investigates urban tree mortality across 959,466 trees in four German cities from 2016 to 2022 and identified key factors affecting tree mortality. While assessing annual mortality rates, we found an average annual mortality rate of 1.3 %, with a peak in 2019. Street trees exhibited a 0.3 % higher mortality rate than non-street trees, underscoring their vulnerability to urban conditions. Additionally, annual mortality rates were higher for coniferous species (2.2 %) compared to broad-leaved species (1.3 %). Using multivariable logistic regression approach, significant correlations were found between mortality probability, drought tolerance indices, and local climate determined by de Martonne Index (dMI). Our findings showed that younger and drought-intolerant street trees are likely more vulnerable to urban environmental stressors than matured, drought-tolerant ones. The study successfully assessed cadaster data, demonstrating their utility for urban tree mortality monitoring but highlighted challenges such as difficulty in accessing data, inconsistent data collection intervals, various data formats and less frequent monitoring of non-street trees. This research underscores the need for long-term monitoring and highlights the importance of species-specific and environmental factors in urban forestry management.

1. Introduction

Urbanization threatens natural ecosystems, impacting ecological patterns like species richness and carbon sequestration, consequently, the ecosystem service (ESS) provisions (Allen et al., 2017; Nowak & Crane, 2002). Urban greening initiatives, with trees playing a pivotal role, present a viable solution to mitigate these threats (McPherson et al., 2007; Moser et al., 2017). However, urban trees in particular street trees face severe survival risks due to the changing climate (Czaja et al., 2020). The loss of a healthy tree not only declines ecological benefits but also causes substantial financial costs for replacement. In Germany, the total cost of planting a new street tree, including acquisition, planting, and site preparation, ranges from €500 to €2400, with an average of €1450 per tree (Pauleit et al., 2002). Early-stage mortality, which is often highest within the first five years after planting, substantially increases the need for replanting and the associated financial

costs (Nowak & Aevermann, 2019). Consequently, there has been a growing emphasis in the last decade on tree monitoring (Croeser et al., 2020; Doll et al., 2018; Lizée et al., 2011; May et al., 2013; Petrova et al., 2025), management and planning (Breger et al., 2019; Gómez-Baggethun & Barton, 2013; Ignatieva et al., 2011) and urban environment and ecosystems (Livesley et al., 2016; Moser et al., 2015; Rahman et al., 2020; Rötzer et al., 2019). Nonetheless, there is still comparatively less literature specifically addressing tree mortality in urban environments. Numerous studies have explored tree mortality in forest environments (e.g. Hartmann et al., 2015; Pretzsch et al., 2023a; Pretzsch et al., 2023b). However, tree mortality in forests is mainly driven by intra- or inter-specific competition for resources, which involves the elimination of trees (Pretzsch et al. 2023a). Urban areas, in contrast, are characterized by complex social-ecological systems where tree mortality is influenced by a wider range of factors, including human activities and environmental conditions (Esperon-Rodriguez et al.,

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2023; Mincey et al., 2013; Vogt et al., 2015). Thus, findings from forest studies may not be directly applicable to urban trees, highlighting a significant knowledge gap.

Hilbert et al. (2019), noted that studies in forest ecology report annual mortality rates (AMR) of 1–3 % or lower for mature canopy trees (Condit et al., 1995; Harcombe and Marks, 1983; Lorimer et al., 2001). In urban environments, however, tree mortality rates vary widely based on factors like the growing environment (e.g. streets or parks) and monitoring methods (e.g. repeated inventory of uneven-aged trees or planting cohort of even-aged trees). Repeated inventory studies on uneven aged trees showed varying mortality rates for uneven-aged urban trees: 6.6 % for various urban trees (Nowak et al., 2004); 0.49 % and 1.9 % for street trees with or without stewardship (Boyce, 2010); 3.7 % for street trees (Roman et al., 2014); 3.6 % for all urban trees (Boukili et al., 2017); 2.6–3 % for yard, street and public trees (Steenberg et al., 2018); and 3.06 % versus 1.41 % for street trees and rural trees (Smith et al., 2019). Studies on even-aged planting cohorts also showed diverse AMR: 6 % for streets, parks and schoolyard trees (Thompson et al., 2004); 1.3–3.3 % for various urban trees (Koeser et al., 2014); 2.4–11.2 % for street trees (Roman & Scatena, 2011); 3.1–4.6 % for street, parks and yard trees (McPherson, 2014) and 1.25–4.17 % for street tree pits with and without stewardship (Boyce, 2010).

Many studies indicated that urban tree mortality is influenced not only by severe site conditions but also by tree size and physiological characteristics. Hilbert et al., (2019) found that mortality varies with tree size development, and assessed it using survivorship and mortality curves. These curves function as demographic models, simulating population changes over time through additions and subtractions of individuals (Roman et al., 2016). Nowak et al. (2004) worked on urban trees in Baltimore and identified a U-shaped mortality curve, with highest mortality for smallest and largest tree size classes, and lower mortality in the middle. Conversely, Roman et al. (2014) found the highest mortality probability only in the smallest trees which is known as Type III mortality curve (see the glossary in suppl.).

Additionally, some studies tried to investigate the mortality related to physiological characteristics. Some showed that broad-leaved trees may exhibit higher mortality rates compared to coniferous trees due to their greater water demand, making them more susceptible to urban stressors (Roloff et al., 2009; Taccoen et al., 2022). While another showed that coniferous species like spruce are more sensitive to drought stress than broad-leaved species like beech (Gebhardt et al., 2023). This could relate to species-specific drought tolerance. Niinemets and Valladares (2006) studied 806 tree and shrub species and defined five indexes for drought tolerance (1, very intolerant; 2, intolerant; 3, moderately tolerant; 4, tolerant; 5, very tolerant). They showed that ecological characteristics significantly influence drought tolerance and tree growth. Species with high drought tolerance index (DTI), such as *Robinia pseudoacacia* and *Pinus sylvestris* (DTI: 4.11 and 4.34), are known to be well-equipped to handle water scarcity due to deep rooting systems and efficient water use strategies. Conversely, species with low drought tolerance, such as *Betula pendula* and *Picea abies* (DTI: 1.85 and DTI: 1.75 resp.), are known to be more vulnerable to hydraulic failure and carbon starvation during dry periods (Niinemets and Valladares, 2006). In urban environments, where water availability and growing conditions are often restricted, selecting species with appropriate drought tolerance is essential for reducing tree mortality and ensuring long-term survival (Gillner et al., 2013, 2015).

Local climate is an affecting factor on urban tree growth, as it directly influences tree health through temperature, precipitation patterns, and the occurrence of extreme weather events (Pretzsch et al., 2017). Trees in urban environments are subjected to varying microclimates, which influence their growth rates, and vulnerability to stress. A useful metric for evaluating these climatic variations is the de Martonne Index (dMI), which combines air temperature and precipitation data to determine the level of aridity or humidity (De Martonne, 1926). Dervishi et al. (2022) studied 1178 trees across seven large cities worldwide and found that

trees in more humid cities (higher dMI) exhibit higher growth rates compared to those in drier cities (lower dMI).

Previous studies have addressed the size-based mortality of urban trees but less is known about interaction between tree species and local climate conditions with tree size. This gap presents a critical challenge in effectively managing urban forests. Using cadaster data this study employs a novel approach to understand how environmental conditions, tree species, drought tolerance, and urban climate conditions (dMI) influence urban tree mortality rates and demonstrate the potential of urban tree cadasters as tools for actionable knowledge.

Analyzing tree cadaster data from the four German cities Berlin, Bonn, Erlangen, Fürth using multivariable logistic regression models, we aimed to answer the following questions:

Q1: How do AMR differ among the four cities?

Q2: To what extent does AMR vary based on: a) tree species, and b) tree growing environments?

Q3: How do tree characteristics (size, species, and drought tolerance) influence the mortality probability of street trees?

Q4: Does variation in local climate conditions correspond to differences in the mortality probability of street trees?

2. Material and method

2.1. Study area

This study was conducted in the four German cities Berlin, Bonn, Erlangen, Fürth (Table 1) which were selected based on the availability of detailed urban tree inventory data. These cities also differ in population size and regional climate. Berlin, as the largest city (3.6 million), contrasts with the smaller cities of Bonn (330,000), Erlangen (116,000), and Fürth (130,000) (CityPopulation, 2022). According to the dMI (Pellicone et al., 2019), Berlin is classified as moderately arid, while the others are slightly arid. This diversity allows for broader insights into context-dependent patterns of urban tree mortality and answering our research questions.

2.2. de martonne index calculation

The de Martonne aridity index (dMI), serves as the metric for describing aridity, a prolonged climatic phenomenon characterized by a deficiency in water supply or precipitation (De Martonne, 1926). This index combines the impact of temperature and precipitation by Eq. 1.

$$dMI = \text{Perc} / (\text{AAT} + 10) \quad (1)$$

Here, Perc represents the total annual precipitation (in mm), and AAT (in °C) signifies the long-term mean temperature (2016–2022) per city. Higher dMI values point to more humid climatic conditions, whereas lower values indicate more arid climatic conditions. As shown in Table 1, Berlin stood out with the lowest dMI at 28.0 categorized as moderately arid climate. Following, Erlangen (dMI = 34.6), Fürth (dMI = 31.2) and Bonn (dMI = 30.4) were all classified under slightly arid conditions (Pellicone et al., 2019).

2.3. Data structure of cadasters

For this study we used cadaster data sets which were provided by the city administrations of the selected cities. Despite the cadaster included data from various periods, we restricted ourselves to the time frame of 2016–2022 across all cities for the sake of consistency and comparability. The studied data was based on a full inventory and not a sample inventory. However, since only dead trees were added to the dataset annually, it inherently consisted of single observations. To facilitate analysis, we merged the annual dead tree lists with the main inventory data, thereby including both alive and dead trees. This integration required the use of a tree identification code, referred to as the tree ID, to

Table 1

Location of the four cities in Germany and averages of annual air temperature (AAT), the sum of precipitation (Perc), and average of de Martonne Index (dMI) for the period 2000–2022 (data source: DWD Climate Data Center).

City	Latitude	Longitude	Area size [km ²]	Elevation [m]	AAT [°C]	Perc [mm/year]	dMI
Berlin	52.52	13.40	891.3	40	10.8	580	28.0
Bonn	50.73	7.09	141.06	60	11.3	645	30.4
Erlangen	49.58	11.01	76.95	279	10.0	688	34.6
Fürth	49.46	11.00	63.35	294	9.9	619	31.2



trace individual trees. The cadaster datasets included a broad range of tree attributes. Most important for our study were tree ID, tree species, stem circumference, planting date and felling date, urban setting (street or open area/park trees). Within the cadaster, a distinct column indicated the status of each tree whether it was alive (0) or dead (1). This binary classification served as the fundamental basis for all subsequent analyses. To merge the datasets from each city, we first applied a standardized pre-processing pipeline. This included harmonizing tree species names, excluding invalid diameter measurements, removing duplicate tree IDs, and adding new attributes such as DTI and the dMI for each city and tree species. Only after completing these steps, we start with preparing the data set for AMR calculations and prediction models.

Using Berlin as an example, the dead trees were recorded individually by the Berlin district authorities and reported to Berlin “Green Space Information System”. The cadaster was updated annually in February, during which dead trees were removed from the cadaster list. For this study, we used the most recent update from February 2023, which included all trees still standing as of the end of 2022. To track tree mortality, we received a separate dataset listing all dead trees recorded between 01.01.2016, and 31.12.2022, including the dates when each tree was marked as dead or removed. We merged these records with the February 2023 cadaster data to comprehensively analyze both living and dead trees across the study period between 2016 and 2022. In Berlin, tree circumference measurements were normally updated every two to four years, although this was done inconsistently. As a result, each tree had a single observation indicating whether it was alive or dead, when it was planted and all other tree attributes mentioned above. The final updated tree cadaster dataset, excluding dead trees, is published annually on December 31 and made available digitally on FIS-Broker (<https://fbinter.stadt-berlin.de/>).

In Bonn, Erlangen, and Fürth, the cadaster datasets contained similar variables to those in Berlin, including mentioned tree attributes. While update procedures varied slightly between cities (e.g., annual upload dates), data collection was guided by shared national frameworks such as the GALK standards (e.g. GALK, 2021) and the FLL tree control guidelines (e.g. FLL, 2020). These define inspection frequency based on tree age, vitality, and site-specific risk. This harmonized basis allowed us to compile a consistent and comparable dataset across all cities.

2.4. Data processing for annual mortality calculation

As the first objective of this study, we aimed to explore the AMR in each city. In this study, the data analyzed often lacked information on

the specific reasons for tree death or removal. Therefore, tree mortality was defined as any case where a tree was either observed dead or marked as removed, encompassing all cases of tree loss. Table 2 provides an overview of the data set. The prepared table includes the number of alive, dead, newly planted and total tree number per year and city. To ensure the accuracy of the mortality rates, we minimized data filtration and excluded only incorrect tree species names. As explained above, given the single-observation nature of the cadasters, we did not have annual data on the number of alive trees. Therefore, our initial step was to calculate the number of alive trees for each year in each city using the newly planted and dead tree number attributes. This enabled us to calculate the total number of trees (both alive and dead) per year. To

Table 2

Total number of trees recorded in the cadaster data from 2016 to 2022.

City	Alive	Dead	Planted	Total
Berlin				
2016	909,255	11,275	2838	920,530
2017	898,781	13,670	3196	912,451
2018	886,548	16,068	3835	902,616
2019	875,085	15,666	4203	890,751
2020	862,777	16,822	4514	879,599
2021	850,674	16,230	4127	866,904
2022	834,949	18,841	3116	853,790
Bonn				
2016	70,988	580	331	71,568
2017	70,545	734	291	71,279
2018	69,820	899	174	70,719
2019	69,004	1262	446	70,266
2020	68,488	1457	941	69,945
2021	67,896	1026	434	68,922
2022	67,776	947	827	68,723
Erlangen				
2016	21,353	165	293	21,518
2017	21,193	325	165	21,518
2018	21,454	153	414	21,607
2019	21,469	208	223	21,677
2020	21,753	129	413	21,882
2021	21,951	203	401	22,154
2022	21,973	117	139	22,090
Fürth				
2016	15,185	95	215	15,280
2017	15,193	104	112	15,297
2018	15,408	111	326	15,519
2019	15,271	315	178	15,586
2020	15,116	225	70	15,341
2021	14,863	253	-	15,116
2022	14,733	130	-	14,863

determine the number of living trees in the first year ($i = 1$) of cadaster data, we applied Eq. 2:

$$\text{alive trees}(y_1) = \text{total trees}(y_n) - \text{dead trees}(y_1) - \sum_{i=2}^n (\text{planted trees}(y_i)) \quad (2)$$

Here y_i refers to the very first year of the cadaster's introduction (before 2016) in each city respectively, y_n represents the final year within the cadaster time frame (2022). $\text{total trees}(y_n)$ is the total number of trees (alive and dead) in the last year, $\text{dead trees}(y_1)$ is the number of dead trees recorded in the first year, $\sum_{i=2}^n (\text{planted trees}(y_i))$ is the sum of all newly planted trees from year two to the last year. Following the first year, for each of the subsequent years, we utilized Eq. 3 to calculate their number of alive trees by adding the trees planted and subtracting trees that died over the year in question:

$$\text{alive trees}(y_i) = \text{alive trees}(y_{i-1}) - \text{dead trees}(y_i) + \text{planted trees}(y_i) \quad (3)$$

In Eq. 3, y_i represents the years after the first year, y_{i-1} indicates the year preceding y_i .

As a result, in the latest defined cadastral data (2022), we encompassed a total number of 959,466 trees. Berlin had the highest tree numbers at 853,790, followed by Bonn with 68,723, Erlangen with 22,090, and Fürth at 14,863.

2.5. Annual mortality rate calculation

To answer the first research question, we calculated AMR for each city applying Eq. 4.

$$\text{AMR}(y_n) = \left(\frac{\text{Total number of dead trees}(y_n)}{\text{Total number of tree population}(y_n)} \right) \quad (4)$$

Here y_n represents each year in the studied timeframe. To calculate the AMR of street/non-street and broad-leaved/coniferous tree species we used the same equation.

2.6. Data processing for statistical models

To prepare the data for answering the second research question, we implemented a two-step preprocessing procedure. First, we converted the data from single observations to duplicated entries, allowing us to monitor the status of each tree (alive or dead) over the whole study period (2016–2022). During this stage, alive trees were duplicated and kept as alive, while dead trees were duplicated as alive until their fell date. To study mortality by tree size, we used tree diameter as a co-variable. However, diameter measurements were often unavailable or inconsistent, especially for non-street and alive trees. Thus, to improve model accuracy, we focused on street trees, where data quality was better. While diameter data for dead street trees were reliable, those for living trees were often estimated or irregularly remeasured. Therefore, we weighed models based on the proportion of alive and dead trees with valid diameter measurements (see Table S1). During the filtering process, we excluded invalid diameter and the ones below 2.5 cm, as advised by city registry officials who specified this as the minimum diameter for newly planted trees. We also applied an upper threshold of 100 cm to represent mature tree size in the city.

Second, we assigned drought tolerance index (DTI) from Niinemets and Valladares, (2006) to each tree species to further use it in the models. As an example, we summarized ten most broad-leaved and coniferous tree species in our single observatory data set of street trees in Table 3. As shown, the coniferous species represented only a very small share of the total sample, reflecting commonly reported urban street tree populations in Central Europe, including Germany.

2.6.1. Statistical Models

As our approach for statistically modelling tree mortality probability, we chose mixed logistic regression models. As stated by Salas-Eljatib and

Table 3

Overview of the ten most common tree species in the analyzed data, with their scientific and English names, sample number, drought tolerance indexes (DTI) and drought tolerance class (DTC) from Niinemets and Valladares (2006).

Sc. name	Eng. name	n	Species class	DTI	DTC
<i>Tilia cordata</i>	Small-leaved lime	80,977	Broad leaved	2.75	2.5
<i>Acer platanoides</i>	Norway maple	71,410	Broad leaved	2.73	2.5
<i>Quercus robur</i>	English oak	29,591	Broad leaved	3.35	3.5
<i>Platanus acerifolia</i>	London plane	28,992	Broad leaved	2.95	2.5
<i>Tilia platyphyllos</i>	Large-leaved lime	20,340	Broad leaved	2.52	2.5
<i>Aesculus hippocastanum</i>	Horse chestnut	18,229	Broad leaved	2.82	2.5
<i>Betula pendula</i>	Silver birch	16,177	Broad leaved	1.85	1.5
<i>Acer pseudoplatanus</i>	Sycamore maple	16,112	Broad leaved	2.75	2.5
<i>Robinia pseudoacacia</i>	Black locust	14,558	Broad leaved	4.11	4.5
<i>Fraxinus excelsior</i>	European ash	13,679	Broad leaved	2.5	2.5
<i>Pinus sylvestris</i>	Scots Pine	4032	Coniferous	4.34	4.5
<i>Pinus nigra</i>	Black Pine	638	Coniferous	4.38	4.5
<i>Picea pungens</i>	Blue Spruce	461	Coniferous	2.88	2.5
<i>Picea abies</i>	Norway Spruce	316	Coniferous	1.75	1.5
<i>Larix decidua</i>	European Larch	274	Coniferous	2.31	2.5
<i>Picea omorika</i>	Serbian Spruce	183	Coniferous	2.75	2.5
<i>Chamaecyparis lawsoniana</i>	Port Orford Cedar	72	Coniferous	2.06	2.5
<i>Pinus mugo</i>	Mountain Pine	69	Coniferous	4.23	4.5
<i>Thuja occidentalis</i>	Northern White Cedar	53	Coniferous	2.71	2.5
<i>Pinus strobus</i>	Eastern White Pine	52	Coniferous	2.29	2.5

Weiskittel (2020), logistic regression is the most widely used statistical model for studying tree- and stand-level mortality (Cailleret et al., 2016; Groom et al., 2012; Hülsmann et al., 2017; Young et al., 2017). Moreover, logistic regression model allowed the inclusion of multiple predictors and their interactions, while accounting for data structure and non-independence through random effects (e.g., city, site, climate). Given our focus this model provided a suitable framework for addressing our research questions. By standard, logistic regression uses the logit link function (Eq. 5):

$$l(p) = \ln\left(\frac{p}{1-p}\right) \quad (5)$$

for transforming a probability p , in our case the probability of a tree to die, into a quantity that can be predicted with a linear model. As suggested by Zuur et al. (2009), as the representations of the mortality status 1 and 0 (dead, alive) were strongly unbalanced, we used the complementary log-log link function (Eq. 6):

$$l(p) = \ln[-\ln(1-p)] \quad (6)$$

We fitted the following models to the single tree mortality data:

$$l(p_{ijk}) = \beta_0 + \beta_1 \bullet \text{diameter}_{ijk} + b_i + b_{ij} \quad (7)$$

$$l(p_{ijk}) = \beta_0 + \beta_1 \bullet \text{diameter}_{ijk} + \beta_2 \bullet \text{conifer}_{ij} + b_i + b_{ij} \quad (8)$$

$$l(p_{ijk}) = \beta_0 + \beta_1 \bullet \text{diameter}_{ijk} + \beta_2 \bullet \text{drtol}_{ij} + b_i + b_{ij} \quad (9)$$

$$l(p_{ijk}) = \beta_0 + \beta_1 \bullet \text{diameter}_{ijk} + \beta_2 \bullet \text{dMI}_{ijk} + b_i + b_{ij} \quad (10)$$

In these models, the indexes i , j , and k refer to the levels city, tree ID, and observation, respectively. All models contained the diameter to cover the potential size dependency of mortality. The first model (Eq. 7) only contains the stem diameter as a predictor, while the second model (Eq. 8) uses the dummy variable conifer for contrasting the species groups broad-leaved (coded as 0) and coniferous (coded as 1) in addition. The third model (Eq. 9) uses species groups by DTI (see 2.6) instead. To this end, we incorporated the DTI of each tree species as a continuous variable. The DTI ($drtol_{ij}$) represents each species' resistance to drought, ranged from 1 (highly intolerant) to 5 (highly tolerant). The fourth model (Eq. 10) consists of the annual dMI (see 2.2) in addition to the stem diameter. In this study, we integrated the dMI values from the previous year to the felling year. This approach accounts for the delayed effects of climate conditions, particularly drought, on tree growth and mortality. So far, we have described the fixed effects in the models which are connected to the coefficients $\beta_0, \dots, \beta_2$ to be estimated. In addition, b_i and b_{ij} represented the random effects at the levels of city and tree ID respectively. These were introduced in order to account for the hierarchical structure of the data and the correlations resulting from that structure. We ran the models using R, version 4.2.2, (R Core Team, 2022) and the lme4 package (Bates et al., 2015) for model fitting. To visualize the model results, we used the ggplot2 package (Wickham, 2016).

3. Results

3.1. Annual mortality rates across cities

As the first objective of this study, we calculated the AMR across four cities. An overview of the mortality rates is illustrated in Fig. 1 (Table S2). The average AMR in the four cities during the study period (2016–2022), calculated after Eq. 2, was 1.3 %. Berlin had the highest mortality rate (1.8 %), followed by Bonn (1.4 %), Fürth (1.1 %), and Erlangen (0.9 %). While Berlin showed a stable increasing rate between 2018–2022, peaks occurred in Bonn (2.1 %) in 2020, Fürth (2.0 %) in 2019 and Erlangen (1.5 %) in 2017.

3.2. Influence of tree species and growing environment on annual mortality rates

3.2.1. Mortality rates of broad-leaved vs. coniferous species

To address the second research question, we calculated the AMR of broad-leaved and coniferous species groups (Fig. 2 and Table S3). AMR varied across cities and years, with coniferous species consistently showing higher average mortality (2.2 %) than broad-leaved species (1.3 %). Coniferous mortality peaked in 2019, reaching 3.8 % on average, with particularly high rates in Fürth (10.8 %) and Bonn (3 %). Although it declined in subsequent years, it remained above that of broad-leaved species, for example in 2020 (Fürth: 4.2 %; Bonn: 8 %) and

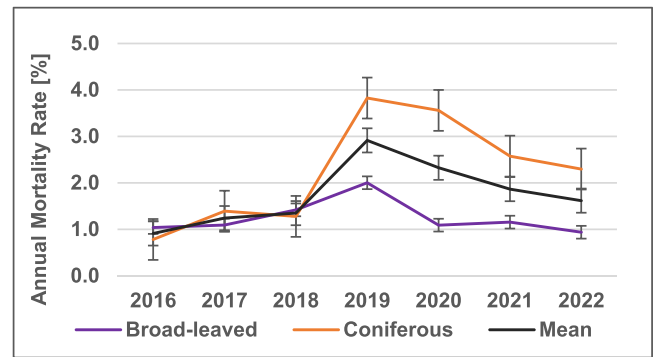


Fig. 2. AMR of broad-leaved and conifer species from 2016 to 2022. The graph displays the mean AMR for broad-leaved species (purple line), conifer species (orange line), and the overall mean (black line). Error bars represent the standard errors of the means. The highest AMR is observed in 2019 for all categories, particularly in coniferous species.

2021 (Berlin: 3.3 %).

In contrast, broad-leaved species displayed relatively stable mortality rates, generally near or below 2.0 %, with only a slight increase in 2019 (Bonn: 1.7 %; Erlangen: 0.9 %). By 2022, broad-leaved mortality declined again, while coniferous mortality, though reduced, remained elevated in several cities (e.g., Fürth: 3.2 %; Berlin: 5.2 %) as shown in Table S3.

3.2.2. Mortality rates in street vs. non-street trees

As a part of the second research question, we calculated the AMR of street and non-street trees. Fig. 3 shows the average mortality rates for all four cities from 2016 to 2022. Street trees consistently exhibited slightly higher mortality rates (1.4 %) than non-street trees (1.1 %). While non-street trees showed a peak in 2019 (1.4 %), street trees showed two rapid peaks one in 2017 (1.5 %) and another in 2019 (1.8 %). Noteworthy differences were observed within each city (Table S4). While Berlin and Bonn had slightly higher mortality rates in non-street (on average 0.6 % and 0.4 % resp.), Erlangen and Fürth showed higher mortality rates in street trees (on average 1.6 % and 0.6 % resp.).

3.3. Impact of tree characteristics on mortality probability

To answer the third research question, we tested various tree traits variables in the logistic regression models on the street trees samples and chose the most accurate ones. The following sections show the result of the models.

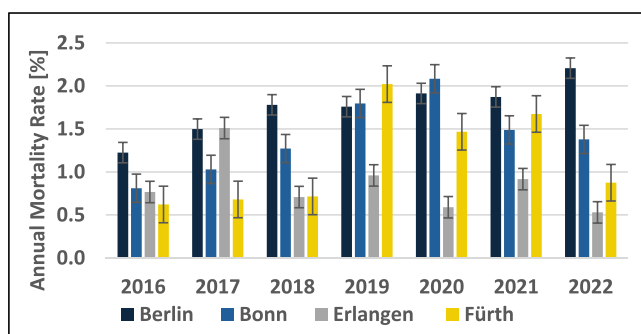


Fig. 1. AMR of the total urban tree population in Berlin, Bonn, Erlangen and Fürth from 2016 to 2022. The bars represent the AMR for each city, with standard error indicated by the error bars.

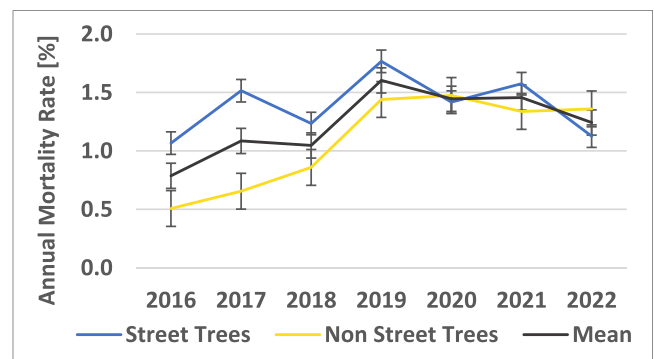


Fig. 3. AMR of street and non-street trees from 2016 to 2022. Mean AMR is shown for street trees (blue), non-street trees (yellow), and average (black). Error bars represent standard errors. A notable peak in AMR is observed in 2019 for both categories.

3.3.1. Tree size development

Fig. 4 shows the probability of mortality as a function of diameter with a continuous decrease of mortality by increase of the tree size, known as Type III mortality pattern. Model 1 after Eq. 7 revealed a significant correlation between diameter and mortality (see Table S7). Result showed, a tree with a diameter of 10 cm, had mortality probability approximately 1.3 %. In contrast, a tree with a diameter of 100 cm had a mortality probability of approximately 0.7 %.

3.3.2. Tree species

Fig. 5 visualizes the result of Model 2 after Eq. 8. We found a significant ($p < 2e^{-16}$) correlation for both broad-leaf and coniferous (Table S7). However, coniferous tree species showed an average 1.2 % higher probability of mortality over the studied diameter range. The result showed, at 10 cm diameter, coniferous trees had 1.5 % higher mortality probability than broad-leaved trees (2.8 % vs. 1.3 %), while at 100 cm diameter, this difference decreased, and coniferous had 0.8 % higher mortality probability than broad-leaved trees (1.5 % vs. 0.7 %).

3.3.3. Tree drought tolerance

Fig. 6 shows the result of Model 3, after Eq. 9. The result revealed a significant correlation between DTI and mortality probability ($p < 2e^{-16}$), indicated that trees with lower DTI are on average 0.7 % more probable to die in comparison to higher DTI (Table S7). For instance, *Betula pendula* with a DTI = 1.85 is expected to have higher mortality probabilities compared to *Robinia pseudoacacia* with a DTI = 4.11. Specifically, for a diameter of 10 cm, the difference in mortality probability between DTI 1.5 and 4.5 is approximately 0.9 % higher, and for a diameter of 100 cm, the difference is around 0.5 % higher.

3.4. Effect of local climate conditions on mortality probability

The results for our fourth research question are shown in Fig. 7, based on Model 4 and Eq. 10. A significant negative correlation ($p = 0.0109$) was identified between mortality probability and dMI (Table S7). However, the model did not indicate notable differences in mortality probability across the dMI ranges of the studied cities. Only a slight variation was observed between dMI 28 (Berlin) and dMI 35 (Erlangen).

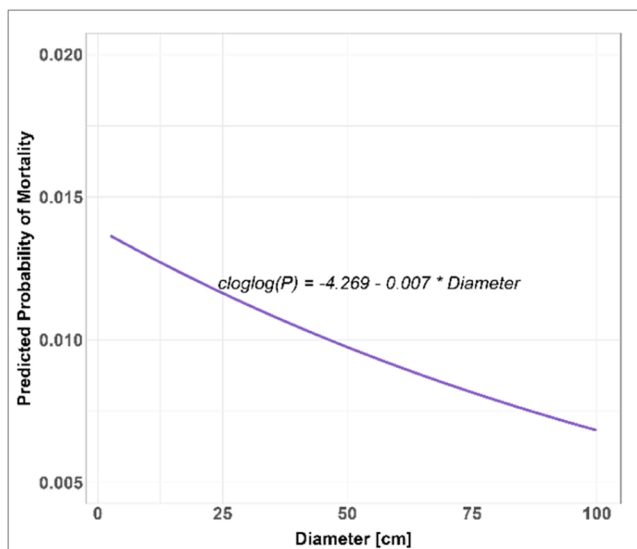


Fig. 4. Probability of street tree mortality as a function of diameter. The model indicates that as the diameter of the tree increases, the probability of mortality decreases.

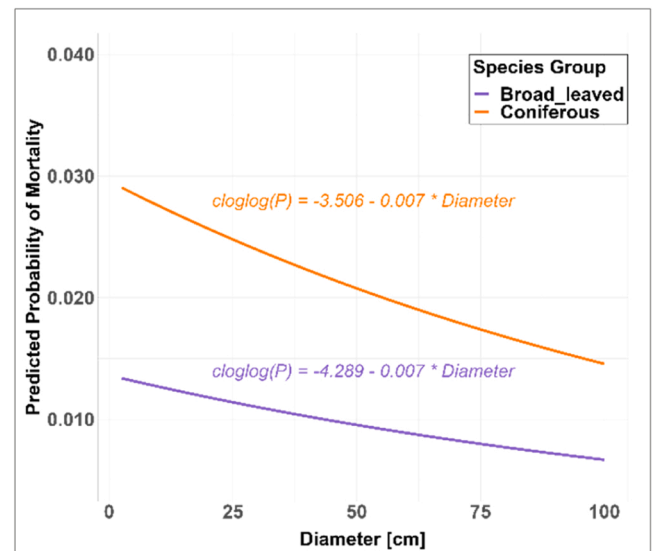


Fig. 5. Probability of street tree mortality by tree species and diameter. The model shows the probability of mortality decreasing with increasing diameter for broad-leaved species (purple) and coniferous species (orange).

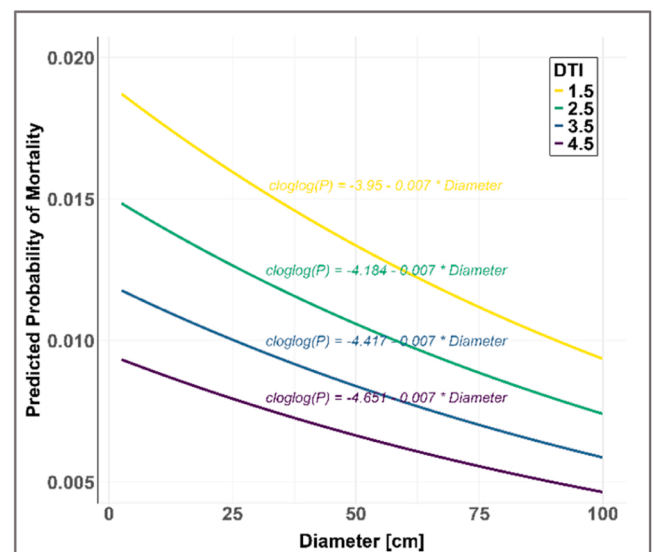


Fig. 6. Model (3): Probability of street tree mortality by DTI (drought tolerance index) and diameter. The model shows the probability of mortality decreasing with increasing diameter for different levels of drought tolerance. The lines represent different DTI means: 1.5 (yellow), 2.5 (green), 3.5 (blue), and 4.5 (purple). The line equations are represented by the corresponding colors.

4. Discussion

Our analysis of urban tree mortality in four German cities revealed critical insights into the factors influencing tree mortality rates. Further by examining mortality probabilities, we identified key factors and correlations between urban settings, tree physiological characteristics, and local climate conditions. The results emphasize how different factors such as tree species, drought tolerance, and environmental conditions shape street tree mortality.

4.1. Impact of tree species and growing environment

Our results showed on average a 1.3 % AMR over all urban tree populations in the four cities during the study period (2016–2022). We

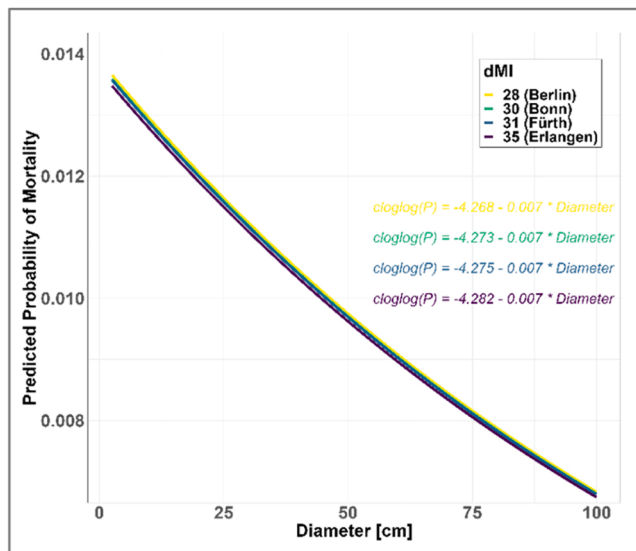


Fig. 7. Probability of street tree mortality by dMI and diameter. The model shows the probability of mortality decreasing with increasing diameter for different levels of the dMI. The lines represent different dMI values: 28 (yellow), moderately arid), 30 (green), 31 (blue), 35 (purple), slightly-arid.

found a relatively low rate compared to similar studies with repeated uneven aged inventories. For instance, Nowak et al. (2004) reported 6.6 % and Boukili et al. (2017) 3.6 % MR for urban trees in the U.S. Steenberg et al. (2018) found rates of 2.6 %-3 % in Canada. When comparing mortality between street and non-street trees, we found consistent higher AMR in street trees (1.4 % vs. 1.1 %), aligning with previous studies that highlighted the adverse conditions faced by trees in urban settings, such as limited root space, soil compaction, and human activities (Roman and Scatena, 2011; Smith et al., 2019). Comparing our result with similar studies on street trees, our AMR in various years (0.9–2.5 %) was aligned with the study of Boyce, (2010) who found AMR between 0.5 % and 1.9 % while studying street trees in pits with and without stewardship for four years in the U.S. However, other studies reported higher MR, for instance, Roman et al., (2014) found a 3.7 % MR, similarly, Smith et al. (2019) found a MR of 3.1 % for street trees in the U.S. The variability in reported AMR both for all urban trees and street trees could arise from differences in calculation methods, extreme climate conditions in certain studies, or structural variations in the data. Additionally, it could point to the possibility that urban tree management strategies in German cities are more effective in sustaining tree populations.

Our results highlight important practical implications, including the need for more frequent monitoring, targeted maintenance, and species selection adapted to street conditions. Higher mortality of street trees also increases replacement costs and potential safety hazards, underscoring the importance of proactive management.

We also found that urban environments cause unique challenges for tree survival, as evidenced by the varying AMR across the four cities (Berlin: 1.8 %, Bonn: 1.4 %, Fürth: 1.2 % and Erlangen: 0.9 %). Differences between cities can reflect variation in regional climate, urban heat island intensity, and sealing degree. Larger cities like Berlin with denser infrastructure and traffic may place greater stress on trees. This aligns with Pretzsch et al. (2017), who found accelerated urban tree growth linked to climate and urban conditions. Management practices, planting rates, and species choice may also contribute to city-level differences. In our study although the intensity of the annual mortality rate varied during the study period, we found the highest mortality between 2019 and 2020 for all the cities, likely due to accumulated severe drought happened in Germany from 2018 to 2019 (Hartmann et al., 2022).

Our findings revealed that coniferous species had a higher average

AMR of 2.2 %, compared to 1.2 % for broad-leaved species across the urban tree population. While broad-leaved species are often considered more sensitive to abiotic stressors, our study found higher vulnerability in coniferous species. This aligns with Gebhardt et al. (2023) who reported a coniferous species like spruce as more drought-sensitive than a broad-leaved species like beech. Escobedo et al. (2016) similarly found slightly higher mortality rates for coniferous species, with an AMR of 3.3 % compared to 2.9 % for broad-leaved species in Santiago, Chile. The slightly higher mortality rates for coniferous trees in urban areas may be attributed to their sensitivity to prolonged drought stress. Other unexamined factors, such as pests and diseases, compacted soils, and air pollution might also negatively impacted these species.

4.2. Role of tree traits and climate on street trees mortality

This study confirmed a Type III relationship between tree size and mortality probability, demonstrating a significant correlation between tree diameter and mortality probability ($p < 2e^{-16}$). This pattern was consistent with previous studies, which suggested that smaller trees are more vulnerable to urban stressors such as mechanical damage and drought (Marchin et al., 2022; Smith et al., 2019). Additionally, Hilbert et al. (2019) reported that urban trees often exhibit either a U-shaped mortality pattern, with high rates in small and large diameter classes, or a Type III pattern, with high rates mainly in smaller classes. When integrating the tree species group (broad-leaved vs. coniferous) to the diameter model, our logistic regression model showed higher mortality probability of coniferous tree species across all diameter ranges. Possible factors include their higher vulnerability to site-specific conditions. Although coniferous trees cover only about 1 % of the street trees in our dataset, similar to proportions reported for other Central European cities, they should not be overlooked. Coniferous contribute significantly to urban ESS. Pace and Grote (2020) demonstrated their high capacity for removing airborne particulate matter due to their year-round foliage, needle structure, and high surface roughness.

We also found that species' drought tolerance significantly influenced mortality probability. Species with higher drought tolerance indices exhibited lower mortality rates, consistent with adaptive traits such as deeper root systems and efficient water use (Niinemets and Valladares, 2006; Stratópoulos et al., 2019). This result was in line with Poschenrieder et al. (2022) who used this index to model sustainable urban tree management in Germany and found that drought-tolerant species (e.g. *Platanus orientalis* and *Robinia pseudoacacia*) had lower mortality rates during dry periods compared to drought-intolerant species (e.g. *Tilia cordata* and *Aesculus hippocastanum*). Moreover, Nitschke et al. (2017) showed that tree growth in Melbourne was negatively impacted by arid conditions and successive drought years, significantly reducing radial growth.

As the last objective of this study, we also examined the relationship between mortality probabilities and cities local climate conditions. We found a significant correlation between dMI and the mortality probability of street trees. This result was aligned with Dervishi et al. (2022), who showed that tree growth in drier cities with lower dMI values experienced reduced growth rates. Similarly, Moser-Reischl et al., (2018) found that higher dMI values, indicating more favorable water conditions, positively influenced tree growth. However, we did not observe noticeable differences in mortality probabilities across the dMI ranges. This could be attributed to the limited climate variation among the studied cities or the influence of additional factors beyond local climate.

Overall, street tree mortality is influenced not only by climate and tree characteristics but also by management practices, pest outbreaks, and urban form. For example, pest and pathogen infestations and non-biological reasons such as street safety, underground construction caused substantial tree mortality in Germany, affecting species composition and ESS (Pauleit et al., 2002; Raum et al., 2023). The combination of biotic stressors and local management responses can confound

climate- or species-related mortality patterns and should be considered when interpreting our results. The original idea for investigating urban tree mortality was inspired by Pretzsch et al. (2021), who developed a scenario modelling framework to enhance sustainable ESS in urban forestry.

5. Conclusion

Our study successfully harnessed cadaster data, providing solid tree mortality information distinct from forest mortality data. It highlighted challenges associated with lack of data accessibility and emphasized the need for consistent and long-term monitoring of urban trees.

While this study contributes to the urban tree monitoring knowledge gap, it is important to acknowledge certain limitations that may influence the interpretation of the results. This study focused on street trees due to their higher data quality for safety control, which may not fully represent the entire urban tree stock. Analyses were conducted at the broad species level rather than for individual species. The small coniferous sample size, reflecting the naturally low proportion in Central European street tree populations, limits comparability with studies from regions where coniferous are more commonly planted. Using cadaster data presents challenges, including limited availability, restrictions on sharing, and complex data structures. Such datasets often lack auxiliary metadata on precise removal causes, preventing distinction between biological mortality and other removals (e.g., construction, hazards). Future inventories should standardize and systematically record this information. While applied species-level DTI is widely used in similar studies, it was developed for forest contexts and may not fully represent urban stressors. The relatively similar climate conditions among the study cities limited us to detect broader climate–mortality relationships. Including cities with a wider climatic range could address this in future research.

Our findings should be interpreted with caution, as street trees face multiple unmeasured (biological and non-biological) stressors. Differences found in AMR between cities may also reflect variation in species composition, planting pit size, irrigation intensity and frequency during droughts, nursery stock origin, winter road saltating, pollutant loads, and other local climatic or management factors, which were beyond the scope of this study.

Overall, our study highlights a shift in the role of urban tree cadasters from static data repositories to dynamic tools for actionable knowledge. Integrating cadaster data which is originally not designed for scientific use, into research frameworks improves data quality, error control, and the range of monitored variables. This can assist sustainable tree management, creates a feedback loop between monitoring and knowledge-based decision-making. Our mortality findings can further enhance scenario modelling approaches by supplying empirical data for real-world applications.

CRediT authorship contribution statement

Hans Pretzsch: Writing – review & editing, Supervision, Methodology, Conceptualization. **Astrid Moser-Reischl:** Writing – review & editing, Methodology, Conceptualization. **Thomas Rötzer:** Writing – review & editing, Methodology, Conceptualization. **Martin Honold:** Writing – review & editing, Data curation. **Peter Biber:** Writing – review & editing, Methodology, Formal analysis. **Leila Parhizgar:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ufug.2025.129098.

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